A+ MCQs — Scenario Based

1. A PC suddenly shuts down during heavy gaming. The CPU temperature is very high. What should you check first?
A. RAM
B. CPU cooling system
C. Keyboard
D. SATA cable
Answer: B — CPU cooling system

2. A technician installs a CPU but the system does not boot. The CPU was forced into the socket. What is the likely issue?
A. Incorrect CPU alignment
B. Faulty monitor
C. Low RAM speed
D. Wrong SATA cable
Answer: A — Incorrect CPU alignment

3. A computer powers on, but there is no display and the motherboard produces beep codes. Which process is reporting the hardware problem?
A. BIOS/POST
B. DHCP
C. DNS
D. GPU driver
Answer: A — BIOS/POST

4. A technician presses the PC case power button, but nothing happens. Which motherboard connection should be checked first?
A. SATA data
B. Power SW header
C. HDMI
D. CPU fan RGB
Answer: B — Power SW header

5. A new motherboard has been installed. The system needs the main motherboard power connection from the PSU. Which connector is normally used?
A. 4-pin Molex
B. SATA power
C. 24-pin ATX
D. RJ45
Answer: C — 24-pin ATX

6. A PC has enough RAM but becomes unstable because the CPU overheats. Which component helps transfer heat away from the CPU?
A. Heat sink
B. SSD

C. CMOS battery
D. NIC
Answer: A — Heat sink

7. A technician wants to install a SATA hard drive. Which cable supplies electrical power to the drive?
A. SATA power cable
B. VGA cable
C. RJ45 cable
D. SATA data cable
Answer: A — SATA power cable

8. A computer loses power frequently during short power cuts, but the user wants enough time to save work and shut down safely. Which device is appropriate?
A. SMPS
B. UPS
C. Heat sink
D. NIC
Answer: B — UPS

9. A technician needs a device that converts incoming AC power into the DC voltages required by PC components. What should be used?
A. UPS
B. SMPS
C. Router
D. Inverter cable
Answer: B — SMPS

10. A PC contains 8 GB RAM but Windows reports less usable memory. The technician suspects a RAM module problem. What is a useful troubleshooting step?
A. Replace the monitor
B. Test one RAM stick at a time
C. Change the SATA cable
D. Reset the printer
Answer: B — Test one RAM stick at a time

11. A user wants better graphics performance for gaming. The motherboard's integrated graphics are insufficient. What should the technician install?
A. Dedicated GPU
B. Sound card
C. Network card
D. RAM heatsink
Answer: A — Dedicated GPU

12. A technician needs a cooling solution that uses a liquid coolant and radiator to remove CPU heat. Which solution is this?

A. Passive cooling
B. Liquid cooling
C. Thermal RAM
D. SATA cooling
Answer: B — Liquid cooling

13. A printer is powered on but does not print. The technician confirms paper and power are available. What should be checked next?
A. Printer connection/driver
B. CPU socket
C. CPU thermal paste
D. BIOS battery
Answer: A — Printer connection/driver

14. A system starts normally but takes much longer to boot than expected. The technician discovers the system uses older firmware rather than modern firmware. Which comparison is relevant?
A. BIOS vs UEFI
B. RAM vs ROM
C. SATA vs HDMI
D. CPU vs GPU
Answer: A — BIOS vs UEFI

15. A technician is installing a CPU. There is a small triangle marking on both the CPU and socket. What should the technician do?
A. Force the CPU down
B. Align the markings before installation
C. Rotate the CPU randomly
D. Remove the socket
Answer: B — Align the markings before installation

16. A PC suddenly restarts when the CPU is under heavy load. Which component should be suspected first?
A. PSU
B. Keyboard
C. Mouse
D. Monitor
Answer: A — PSU

17. A technician removes the RAM and reseats it because the system is not detecting memory. What type of troubleshooting is this?
A. Replacing the CPU
B. Reseating the DIMM
C. Formatting the SSD
D. Updating the printer driver
Answer: B — Reseating the DIMM

18. A computer powers on, fans spin, but there is no display. POST produces a memory-related beep code. What should be checked first?
A. RAM
B. Printer
C. Ethernet cable
D. Speaker
Answer: A — RAM

19. A user wants temporary memory that the CPU can access quickly while programs are running. Which component provides it?
A. SSD
B. RAM
C. HDD
D. ROM
Answer: B — RAM

20. A CPU is described as having 8 cores and 16 threads. What does this indicate?
A. 16 physical CPUs
B. 8 physical processing cores with 16 execution threads
C. 16 RAM modules
D. 8 hard drives
Answer: B — 8 physical processing cores with 16 execution threads

21. A technician notices the CPU temperature rises rapidly after replacing the CPU. What installation-related item should be checked?
A. Thermal interface material and cooler installation
B. Keyboard driver
C. Ethernet settings
D. Printer spooler
Answer: A — Thermal interface material and cooler installation

22. A PC has integrated graphics but the user installs a separate graphics card for demanding applications. What type of GPU is the new card?
A. Virtual GPU
B. Dedicated GPU
C. Firmware GPU
D. Embedded RAM
Answer: B — Dedicated GPU

23. A technician connects a SATA SSD to a motherboard but the drive has no power. Which connection should be checked?
A. SATA power
B. RJ45
C. HDMI
D. Power SW
Answer: A — SATA power

24. A computer has a working PSU but cannot stay powered during a building power outage. Which device provides temporary backup power?
A. SMPS
B. UPS
C. Heat sink
D. CPU
Answer: B — UPS

25. A technician hears a beep sequence immediately after pressing the power button. Which startup process is responsible for checking hardware before the OS loads?
A. POST
B. DHCP
C. DNS
D. FTP
Answer: A — POST

26. A technician is replacing a CPU. The new processor does not sit correctly in the socket. What should the technician do?
A. Apply force
B. Check the CPU/socket alignment markings
C. Bend the CPU pins
D. Tighten the heatsink first
Answer: B — Check the CPU/socket alignment markings

27. A PC works normally for 10 minutes but shuts down after becoming very hot. Which problem best matches the symptoms?
A. Overheating
B. Incorrect keyboard layout
C. Printer queue
D. DNS failure
Answer: A — Overheating

28. A technician needs to identify whether a PSU can provide power to the motherboard. Which connector is the primary motherboard power connector?
A. 24-pin ATX
B. RJ45
C. SATA data
D. 3.5 mm audio
Answer: A — 24-pin ATX

29. A user presses the power button, but the motherboard does not receive the signal from the case button. Which connection is most relevant?
A. Power SW
B. SATA data
C. USB header

D. Audio header
Answer: A — Power SW

30. A printer is connected but the computer cannot communicate with it because the required software is missing. What should be installed?
A. Printer driver
B. CPU driver
C. BIOS battery
D. RAM driver
Answer: A — Printer driver

31. A technician wants faster boot times and modern firmware features on a new PC. Which firmware technology is generally associated with newer systems?
A. UEFI
B. POST
C. SATA
D. CMOS
Answer: A — UEFI

32. A technician wants to diagnose whether a RAM module or RAM slot is causing a boot problem. Which approach is most useful?
A. Test modules individually and in different slots
B. Replace the monitor
C. Change the keyboard
D. Disconnect the printer
Answer: A — Test modules individually and in different slots

33. A gaming PC becomes unstable only when running graphically intensive applications. The dedicated GPU requires substantial power. Which component should be checked for adequate power delivery?
A. PSU
B. Mouse
C. Keyboard
D. Printer
Answer: A — PSU

34. A technician installs a CPU cooler but forgets the thermal interface material. What is the main concern?
A. Poor heat transfer from CPU to cooler
B. Loss of network connectivity
C. RAM becoming permanently full
D. Printer malfunction
Answer: A — Poor heat transfer from CPU to cooler

35. A user wants protection from brief power interruptions and enough time to safely shut down the PC. Which solution fits the requirement?

A. UPS
B. Heat sink
C. Dedicated GPU
D. SATA cable
Answer: A — UPS